

My AI agent's blast radius

— it doesn't run on my machine —

**A container it can't leave,
and no way out to the internet.**

watch it fail safely →

1 ONE COMMAND

The sandbox is a Dockerfile in the repo. One command builds it — no hand-rolled container setup.

```
• (.venv) natarajan@natarajan-ai-ml-02:~/book04/ai-coding-agent$ ai-agent build-sandbox
Building ai-agent-sandbox:latest from
/home/natarajan/book04/ai-coding-agent/src/agent/sandbox/Dockerfile ...
[+] Building 1.5s (9/9) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile              0.0s
=> => transferring dockerfile: 890B                             0.0s
=> [internal] load metadata for docker.io/library/python:3.12-slim 1.4s
=> [internal] load .dockerignore                                0.0s
=> => transferring context: 2B                                    0.0s
=> [1/5] FROM docker.io/library/python:3.12-slim@sha256:c3d81d25b3154142b0b42eb1e61300024 0.0s
=> CACHED [2/5] RUN apt-get update && apt-get install -y --no-install-recommends 0.0s
=> CACHED [3/5] RUN pip install --no-cache-dir pytest ruff      0.0s
=> CACHED [4/5] RUN useradd --create-home --uid 1000 agent      0.0s
=> CACHED [5/5] WORKDIR /workspace                             0.0s
=> exporting to image                                           0.0s
=> => exporting layers                                           0.0s
=> => writing image sha256:b94803c5933051399a3e4ad3d707a01b228e885fe2e95b2303778175bbca07 0.0s
=> => naming to docker.io/library/ai-agent-sandbox:latest      0.0s
Built ai-agent-sandbox:latest. Use it with: ai-agent run ... --_sandbox docker
```

2 NOT YOUR MACHINE

I asked it who and where it is. It answers with a container ID and the image's user — not my hostname, not my account.

```
tool: run_command  
{"command": "echo \"host=$(hostname) user=$(whoami)\""}  
}
```

```
result: run_command  
[ok]  
host=2b303c8f0afb user=agent
```

3 NO WAY OUT

pip install inside the sandbox can't even resolve a name. The network is off unless I say otherwise.

```
ai-agent – sandboxed, no network

$ ai-agent run "Run this exact shell command and report what happened: pip install requests" \
  -w ~/demo/box --sandbox docker

Starting Docker sandbox from image ai-agent-sandbox:latest

step 1 → run_command {"command": "pip install requests"}
[exit=1]
Defaulting to user installation because normal site-packages is not writeable
WARNING: Retrying (Retry(total=4, ...)) after connection broken by
'NewConnectionError(...: Failed to establish a new connection:
[Errno -3] Temporary failure in name resolution)': /simple/requests/
... 4 more retries, all the same ...
ERROR: Could not find a version that satisfies the requirement requests
ERROR: No matching distribution found for requests

step 2 → run_command {"command": "python -c 'import requests'"}
Traceback (most recent call last):
  File "<string>", line 1, in <module>
ModuleNotFoundError: No module named 'requests'

→ finish "Tried to install requests via pip but encountered network issues."
```

4 MODEL DIES? FINE

I killed the model server mid-run. It retried with backoff, gave up, and reported why — no hang, no crash.

```
Wrote 31 bytes to a.py
[07/15/26 23:03:54] INFO      step 1: write_file ok=True
→ write_file ['path', 'content']
Wrote 30 bytes to b.py
[07/15/26 23:03:55] INFO      step 2: write_file ok=True
→ write_file ['content', 'path']
Wrote 30 bytes to c.py
[07/15/26 23:03:56] INFO      step 3: write_file ok=True
→ write_file ['path', 'content']
Wrote 30 bytes to d.py
[07/15/26 23:03:57] INFO      step 4: write_file ok=True
[07/15/26 23:03:58] WARNING   chat failed (attempt 1/3): Ollama chat request failed: Server
                               disconnected without sending a response. -- retrying in 0.73s
                               WARNING chat failed (attempt 2/3): Ollama chat request failed: All
                               connection attempts failed -- retrying in 1.95s
[07/15/26 23:04:00] ERROR     chat failed after 3 attempts: Ollama chat request failed: All
                               connection attempts failed
Transient error (Ollama chat request failed: All connection attempts failed). Circuit: closed
ERROR     Error in state executing: Ollama chat request failed: All connection
                               attempts failed
INFO      FSM executing --error--> error
```

Part 6 — earlier: “I gave an AI agent shell access” (the guardrails) · “What makes an AI coding agent production-grade?” · “Your AI agent never reads your code” · “My AI agent runs a real language server”

Want to build one?

This entire agent is a book —
and a free, open-source repo.



Get the book (Kindle)

amazon.com/dp/B0H8B6QXXX · amazon.in/dp/B0H8B6QXXX



Clone the code (MIT)

github.com/Natarajan-R/local-ai-coding-agent

Follow for the build, one file at a time.

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